Cashflow Monitor

Product Methodology & Specification

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Author	AFianco Product & Engineering
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1. Executive Summary

AFianco's **Cashflow Monitor** is a real-time financial reporting module designed for Italian small and medium enterprises (SMEs). It transforms raw transactional data - sales, operating expenses, supplier purchases, and fixed costs - into a structured set of KPIs, a composite Financial Health Score, diagnostic indicators, and AI-generated explanations.

The module serves three core purposes:

1. **Operational visibility.** Business owners see daily inflows/outflows, net result, cost structure, and burn rate at a glance.
2. **Financial health assessment.** A composite score (0-100) across five distinct financial dimensions summarizes business health with clear thresholds.
3. **Actionable guidance.** AI-powered explanations translate numeric scores into plain-language assessments, identify priorities, and surface data quality warnings.

Key design principles:

- **No false-good scores.** Missing data dimensions are excluded rather than defaulted to misleadingly favorable values.
- **Scored vs. diagnostic separation.** Only five dimensions drive the score; all others are context-only.
- **Proxy metrics are labeled.** Any approximation (e.g., operational coverage) is explicitly marked and caveated.
- **AI is bounded by data.** The AI reads structured outputs and does not invent financial facts.

2. Intended Audience

Audience	What they should get from this document
Product managers	Complete specification for feature governance, roadmap planning, and QA validation
Engineers	Exact formulas, thresholds, and edge cases for implementation parity
Investors / Board	Methodology credibility, intellectual property clarity, and product maturity signal
Compliance / Legal	Disclaimer boundaries, proxy metric labeling, and AI limitation disclosures
Customer Success	Understanding of what each metric means for end-user support

3. Use Cases

#	Use Case	Actor	Feature
UC-1	Monitor daily revenue vs. expenses	Business owner	KPI cards, daily chart
UC-2	Understand profitability after all costs	Business owner	Net Result, Health Score
UC-3	Check break-even coverage	Business owner	Break-Even card, Structural Strength
UC-4	Identify cash cycle inefficiencies	Finance mgr.	DSO, DPO, Cash Cycle dimension

#	Use Case	Actor	Feature
UC-5	Quick health summary for board	CEO / Owner	Health Score gauge, AI explanation
UC-6	Detect anomalies early	Business owner	Alert rules, Revenue Dynamics
UC-7	Estimate operational runway	Business owner	Operational Coverage (proxy)
UC-8	Assess data completeness	Any user	Confidence, data caveats

4. Purpose of the Cashflow Report

The Cashflow Report provides a structured, automated financial overview that translates raw transaction data into business-relevant metrics. It is not a replacement for professional accounting or CFO-level analysis. It is a **directional tool** that helps business owners understand where money is going, identify early warning signs, track trends over time, and communicate financial posture to stakeholders.

The module distinguishes between **facts** (recorded transactions), **derived metrics** (calculated from facts), and **proxies** (approximations where direct data is unavailable).

5. Scope of the Module

Included: Revenue/expense tracking (daily), four-bucket cost model, profitability KPIs, cash cycle KPIs, operational KPIs, composite health score (0-100), PoP/YoY comparisons, AI explanations, alert rules, payment aging and 60-day forecast.

Not included: Bank account balance integration, tax/VAT filing, inventory/COGS, multi-currency consolidation, intercompany reconciliation, payroll processing.

6. Data Inputs and Source Variables

6.1 Raw Financial Inputs

Input Category	Key Fields	Source
Sales (Revenue)	date, amount, category, due_date, payment_status	CSV/XLSX or manual
Operating Expenses	date, amount, category	CSV/XLSX or manual
Supplier Purchases	date, quantity, unit_price, total_price, iva, total_with_iva, supplier, due_date, payment_status	CSV/XLSX or manual
Fixed Costs	name, category, amount, frequency, start_date, end_date	Manual entry

6.2 Derived Aggregates

Variable	Definition
total_sales	Sum of all sales amounts in the selected period
total_expenses	Sum of all operating expense amounts in the selected period
supplier_purchases	Sum of purchase totals (coalesces total_with_iva, amount, total_price)
fixed_costs_total	Sum of active fixed costs, prorated to period based on frequency
variable_outflows	total_expenses + supplier_purchases
total_outflows	variable_outflows + fixed_costs_total
net_after_fixed	total_sales - total_outflows
open_receivables	Sum of unpaid sales with due_date set
open_payables	Sum of unpaid purchases with due_date set
period_days	(end_date - start_date).days + 1 (calendar days, inclusive)

7. Derived KPI Definitions

All computations use a **canonical formula layer** (kpi_formulas.py) ensuring identical results across the overview, insights, and alert systems.

7.1 Profitability KPIs

Net Margin %

Purpose	Measures percentage of revenue remaining as profit after all costs
Formula	(net_after_fixed / total_sales) x 100
Unit	Percentage (1 decimal)
Returns None	total_sales <= 0
Interpretation	> 15% excellent; 5-15% adequate; < 0% loss
Caveat	If fixed costs not uploaded, margin is artificially inflated

Cost-to-Revenue Ratio

Purpose	Shows how much of every euro of revenue is consumed by costs
Formula	(total_outflows / total_sales) x 100
Unit	Percentage (1 decimal)
Returns None	total_sales <= 0
Interpretation	< 80% healthy; > 100% spending more than earning
Note	Diagnostic only - not scored

7.2 Break-Even KPIs

Break-Even Point

Purpose	Minimum revenue needed to cover all costs
Formula	fixed_costs_total / (1 - variable_cost_ratio)
Unit	EUR (2 decimals)
Returns None	VCR is None, OR fixed_costs <= 0, OR VCR >= 1.0
Caveat	When VCR >= 1.0, break-even is structurally unreachable

Break-Even Headroom %

Purpose	How far above/below break-even the current revenue sits
Formula	((total_sales - break_even) / break_even) x 100
Unit	Percentage (1 decimal)
Interpretation	Positive = safe; Negative = deficit

Fixed Cost Ratio

Purpose	Weight of fixed costs relative to revenue
Formula	(fixed_costs_total / total_sales) x 100
Interpretation	< 15% lean; 15-30% moderate; > 30% heavy

7.3 Cash Flow Timing KPIs

DSO (Days Sales Outstanding)

Formula	(open_receivables / total_sales) x period_days
Unit	Days (1 decimal)
Returns None	total_sales <= 0 OR period_days <= 0
Caveat	Estimated. Accuracy improves with due dates and payment status

DPO (Days Payable Outstanding)

Formula	(open_payables / supplier_purchases) x period_days
Unit	Days (1 decimal)
Returns None	supplier_purchases <= 0 OR period_days <= 0
Caveat	Estimated. Accuracy improves with due dates and payment status

Cash Conversion Gap

Formula	DSO - DPO
Interpretation	Negative = favorable (cash in before costs due); Positive = gap to finance
Returns None	Both DSO and DPO are None
Fallback	When only one computable, missing value treated as 0

7.4 Operational KPIs

Burn Rate (Daily)

Formula	total_outflows / period_days
Unit	EUR/day (2 decimals)
Interpretation	Higher burn rate = faster cash consumption

Operational Coverage (PROXY)

Formula	net_after_fixed / (total_outflows / period_days)
Unit	Days (1 decimal)
WARNING	This is NOT real liquidity. No bank balance data is used.
UI Label	Always displayed as "Operational Coverage ~ estimated"
In Health Score	Diagnostic only - does NOT drive any scored dimension

8. KPI Cards Overview

Detail Tab - Row 1 (Core Operational)

Pos	Card	Value Source	Trend	Color Logic
1	Total Revenue	total_sales	PoP %	Green
2	Operating Expenses	total_expenses	PoP % (inv.)	Red
3	Supplier Purchases	supplier_purchases	-	Red
4	Fixed Costs	fixed_costs_total	-	Default
5	Net Result	net_after_fixed	-	Green >= 0; Red < 0
6	Burn Rate	burn_rate_total	-	Default

Detail Tab - Row 2 (Structural + Diagnostic)

Pos	Card	Color Logic	Styling
1	Break-Even	Green if sales > BE; Red if < BE	Standard
2	Operational Coverage ~ est.	Red < 30d; Green >= 60d	Standard + proxy badge
3	Fixed Cost Ratio	Red > 30%; Green <= 30%	Standard
4	Operating Margin %	-	DIAGNOSTIC (muted, 75% opacity)
5	Outflow Ratio	-	DIAGNOSTIC (muted, 75% opacity)

Cards 4 and 5 use diagnostic styling: reduced opacity, muted border, no success/danger coloring. This visually communicates they are supplementary context, not primary decision metrics.

9. Financial Health Score Methodology

The Financial Health Score is a composite indicator from **0 to 100** that summarizes business health across five financially distinct dimensions. It is designed as a **directional signal**, not a precise diagnosis.

9.1 Design Principles

No redundancy. Each dimension measures a distinct financial concept. Previous v2.x had 8 dimensions with ~90% correlation between 3 pairs.

Not-computable > false-good. Insufficient data excludes the dimension; weight is redistributed.

Scored vs. diagnostic. Only five dimensions drive the score; all others are context-only.

Confidence transparency. A value (0.0-1.0) indicates what fraction of weight was computable.

9.2 Dimension Weights

#	Dimension	Key	Weight	What It Measures
1	Net Margin	net_margin	25 pt	Net profitability after all costs
2	Revenue Dynamics	revenue_dynamics	20 pt	Revenue and margin trajectory
3	Structural Strength	structural_strength	20 pt	Break-even distance + fixed cost leverage
4	Cash Cycle	cash_cycle	25 pt	Cash conversion timing (DSO - DPO)
5	Operational Risk	operational_risk	10 pt	Alert signals + data completeness
			100 pt	Total

9.3 Score Bands

Score Range	Label	Color	Meaning
80-100	Excellent	Green (#22C55E)	Strong financial position across most dimensions
60-79	Good	Yellow (#EAB308)	Generally healthy with areas to monitor
40-59	Attention	Orange (#F97316)	One or more dimensions require action
0-39	Critical	Red (#EF4444)	Significant financial stress detected

9.4 Weight Rescaling

When dimensions are disabled or not computable, remaining weights are **proportionally rescaled** to maintain a 0-100 scale.

```
rescaled_max = (original_weight / computable_weight_sum) x 100
rescaled_points = (original_points / original_max) x rescaled_max
confidence = computable_weight / 100
```

10. Scored Dimensions - Detail

10.1 Net Margin (25 pt)

Formula: margin_pct = (net_after_fixed / total_sales) x 100. Not computable when total_sales <= 0.

Net Margin %	Points (of 25)	Assessment
> 15%	25	Excellent profitability
> 10%	21	Strong
> 5%	16	Adequate
> 0%	10	Marginal
= 0%	5	Break-even
< 0%	0	Loss

10.2 Revenue Dynamics (20 pt)

Composite: Sub-A (Sales Trend, 10 pt) + Sub-B (Margin Trend, 10 pt). If only one computable, it is rescaled to 20 pt.

Sales Trend	Points		Margin Trend (pp)	Points
> +10%	10		> +2 pp	10
> 0%	7		> -2 pp	7
> -10%	4		> -5 pp	4
> -20%	2		<= -5 pp	0
<= -20%	0			

10.3 Structural Strength (20 pt)

Composite: Sub-A (Break-Even Headroom, 10 pt) + Sub-B (Fixed Cost Ratio, 10 pt). Same rescaling logic.

Headroom %	Points		FC Ratio	Points
> 30%	10		< 15%	10
> 15%	7		< 25%	7
> 0%	4		< 35%	4
<= 0%	0		< 50%	2
			>= 50%	0

10.4 Cash Cycle (25 pt)

Based on Cash Conversion Gap (DSO - DPO). Not computable when has_payment_status_data = false.

Cash Conversion Gap (days)	Points (of 25)	Assessment
<= 0	25	Cash arrives before costs are due
< 15	19	Short gap - manageable

Cash Conversion Gap (days)	Points (of 25)	Assessment
< 30	13	Moderate gap
< 60	6	Significant gap - working capital pressure
>= 60	0	Severe gap - financing likely needed

10.5 Operational Risk (10 pt)

Always computable. Sub-A: High alerts (5 pt). Sub-B: Data sources (5 pt).

High Alerts	Points		Data Sources	Points
0	5		>= 4	5
1-2	2		>= 3	3
> 2	0		>= 2	1
			< 2	0

11. Diagnostic Metrics

The following metrics are computed and displayed for context but are **not scored** - they do not affect the composite health score.

Metric	Formula	Unit	Purpose
Cost-to-Revenue Ratio	(total_outflows / total_sales) x 100	%	Overall cost pressure
Operating Margin %	((sales - variable_outflows) / sales) x 100	%	Profitability before fixed costs
Burn Rate (Daily)	total_outflows / period_days	EUR/day	Cash consumption speed
Operational Coverage	net_after_fixed / (total_outflows / period_days)	Days	PROXY: margin-to-burn-rate coverage
DSO	(open_receivables / total_sales) x period_days	Days	Individual collection timing
DPO	(open_payables / supplier_purchases) x period_days	Days	Individual payment timing

Why diagnostic, not scored? These metrics either overlap with scored dimensions (~90% correlated) or are individual components of a scored composite. Scoring them independently would create redundancy and double-counting.

12. Rules for Missing Data / Not-Computable Logic

When a dimension lacks sufficient data, it is marked as **not_computable** rather than defaulted to zero. This prevents artificially favorable (or unfavorable) scores.

Dimension	Trigger	Reason Displayed
Net Margin	total_sales <= 0	No revenue data
Revenue Dynamics	Both trends are None	No previous period data
Structural Strength	Both sub-scores fail	Insufficient data (no fixed costs or revenue)
Cash Cycle	No payment status data or gap = None	No payment status data available
Operational Risk	Never - always computable	-

Sparse-Data False-Good Safeguard

A special guard triggers when only **1 data source** is uploaded AND net margin exceeds **50%** (artificially inflated because no costs are recorded). The system appends a prominent warning in both the structured caveats and the explanation text.

13. Proxy Metrics and Interpretation Caveats

Operational Coverage - Detailed Caveat

Approximates	How long the business can sustain operations
Actually measures	Ratio of net margin to daily burn rate
Does NOT include	Bank balance, credit lines, receivable collection, payable schedules

Why misleading	Business with EUR 50k profit and EUR 5k/day burn shows 10 days but may have EUR 200k in bank
UI Label	"Operational Coverage ~ estimated"
In Health Score	Diagnostic only - does NOT drive any scored dimension
AI Handling	Must qualify with "approximately" or "estimated from burn rate"

14. AI Layer on Top of the Financial Engine

14.1 Architecture

Two modes: (1) **Rule-based explanation** (default, zero API cost, every page load) and (2) **AI-powered explanation** (on-demand via Claude API). Both read the same structured outputs. Neither invents financial facts.

14.2 What the AI Can Answer

Domain	Capability
Profitability	Explain whether the business is profitable, by how much, and trend direction
Cost pressure	Identify which cost bucket is heaviest relative to revenue
Break-even	Explain distance above or below break-even threshold
Trends	Describe revenue and margin trajectory period-over-period
Cash cycle	Explain gap between collection and payment timing
Data completeness	Warn when score is based on limited data; suggest uploads
Priority actions	Suggest most impactful improvement areas from weakest dimensions
Proxy qualification	Clarify when a metric is an estimate, not a measured fact

14.3 What the AI Cannot Reliably Infer

Limitation	Reason
Actual bank balance	No bank account data is ingested
Tax obligations	Tax computation is out of scope
Future projections	Forecasting requires assumptions the system does not make
Industry benchmarks	No sector-specific comparison norms
Root cause analysis	Can identify WHAT is weak but not WHY
Accounting accuracy	Reflects uploaded data quality
Investment advice	Not a financial advisor

14.4 AI Guardrails

- Temperature: **0.2** (conservative, deterministic output)
- Maximum output: **200 tokens**
- Fallback: if Claude API unavailable, **always falls back** to rule-based explanation
- System prompt explicitly instructs to never overstate proxy metrics

15. Example Interpretation Logic

Scenario: Growing E-commerce with Cash Cycle Pressure

Revenue EUR 200k, Margin 20%, Sales Trend +25%, Margin Trend +3pp, Fixed Costs EUR 20k, DSO 55 days, DPO 10 days, 0 high alerts, 4 data sources.

Dimension	Points	Assessment
Net Margin (25)	25/25	20% margin - excellent
Revenue Dynamics (20)	20/20	Strong growth in both sales and margin
Structural Strength (20)	20/20	High headroom, moderate fixed costs
Cash Cycle (25)	6/25	45-day gap - significant WC pressure
Operational Risk (10)	10/10	No alerts, complete data

Score: ~81 (Excellent), Confidence: 1.0. Strongest: Net Margin, Dynamics, Structure. Weakest: Cash Cycle (24% - high-priority issue). Priority Action: Accelerate collections or negotiate longer payment terms.

16. Limitations

Not a bank statement. Operates on uploaded data, not real-time bank feeds.

Garbage in, garbage out. Incomplete/incorrect uploads produce inaccurate KPIs.

No inventory or COGS. Manufacturing/retail should interpret margins accordingly.

Fixed-period comparisons. Seasonal businesses may see misleading trends.

Sector-agnostic thresholds. May be too generous or strict for specific industries.

Not a substitute for professional advice. Consult accountants/CFOs for material decisions.

17. Glossary

Term	Definition
Break-Even Point	Revenue level where total costs equal total revenue
Burn Rate	Average daily total outflows
Cash Conversion Gap	DSO minus DPO; days between paying suppliers and collecting from customers
Confidence	Fraction of total health score weight that was computable (0.0-1.0)
Diagnostic Metric	Metric displayed for context, not included in health score
DPO	Days Payable Outstanding - average days to pay suppliers
DSO	Days Sales Outstanding - average days to collect from customers
Net Margin	Net profit after all costs, as percentage of revenue
Not Computable	Dimension excluded from score due to insufficient data
Operational Coverage	PROXY metric: margin-to-burn-rate coverage in days
Period Days	Calendar days in selected period, inclusive
Proxy Metric	Approximation for a concept without direct data
Rescaling	Redistribution of weight from not-computable dimensions
VCR	Variable Cost Ratio - variable outflows / revenue

18. Governance / Revision Notes

Field	Value
Document Version	v1.0
Status	Draft for Review
Model Version	Health Score v3.0 (5 dimensions)
Previous Model	v2.x (8 dimensions - deprecated)
KPI Formula Layer	Canonical (kpi_formulas.py) - single source of truth
Last Code Audit	March 2026
Test Coverage	722 backend tests passing
Locales Supported	Italian, English, German, French
AI Model	Claude (Anthropic), temperature 0.2, max 200 tokens

Planned for v1.1:

- Confidence badge more prominent in Health Score gauge UI
- Sector-specific threshold profiles (e.g., restaurant vs. manufacturing)
- Bank feed integration to replace proxy operational coverage with real liquidity